Improving Factuality of Large Language Models through Retrieval-Augmented Generation and Self-Verification

NLP Project Proposal — Technical Research Pipeline

~ Vamaxi Maheshwari (230121064), Venkat Sai Teja Kalluri (230121032), Chodabattula Moni Praneeth (230121014)

Core Problem

LLM hallucinations undermine reliability in knowledge-intensive applications

Proposed Solution

RAG + self-verification + claim-level correction pipeline

Research Question

Can combining Retrieval-Augmented Generation, self-consistency, and claim-level verification improve the factuality of LLM-generated responses compared with standard LLM and RAG systems?

Evaluation Scope

6 systems compared across 4 benchmark datasets

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1) Motivation

The Hallucination Problem

Large Language Models can generate fluent and convincing responses that contain unsupported, fabricated, or contradictory information — commonly referred to as **hallucinations**. This limits their reliability in applications such as question answering, education, research assistance, and information retrieval.

Limitations of Existing Approaches

Retrieval-Augmented Generation (RAG) reduces factual errors by supplying external evidence, but **retrieval alone does not guarantee** that the generated response is fully supported by the retrieved information.

Recent work such as **SelfCheckGPT** and **Chain-of-Verification** demonstrates that consistency checking and explicit verification can further improve factuality beyond what retrieval alone achieves.

Proposed Pipeline

Generate an initial answer → detect unsupported claims → retrieve or verify relevant evidence → revise the answer. A practical, experimentally rigorous end-to-end system.

Central Research Question

How much can hallucination detection and self-verification improve factuality compared with a standard LLM and a conventional RAG system, and what is the associated cost in **latency and computation**?

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2) Problem Description

System Goal

Given a user question, an LLM may produce an answer containing a mixture of correct and hallucinated statements. The goal is to build an NLP system that executes the following five-stage objective:

1 Generate

Produce an initial answer to a question

2 Identify

Identify claims that may be unsupported or inconsistent

3 Retrieve

Retrieve relevant evidence or perform independent verification

4 Decide

Assign a hallucination/factuality decision to each claim

5 Correct

Produce a corrected final answer with evidence/citations where possible

Systems to Compare

(i) Standard Prompting Baseline

Vanilla LLM generation with no retrieval or verification

(ii) RAG

Retrieval-Augmented Generation with external evidence supply

(iii) SelfCheckGPT

Self-consistency-based hallucination detection via semantic/NLI agreement

(iv) Chain-of-Verification

Explicit step-by-step verification reasoning over generated claims

(v) Hybrid Pipeline

Detector + evidence retrieval + correction — the full proposed system

Main Hypothesis: Combining retrieval with independent verification and claim-level correction will reduce hallucination rate more reliably than using generation or retrieval alone, while maintaining acceptable answer quality and computational cost.

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3) Dataset Description

HaluEval

35,000 examples — 5K general queries + 30K task-specific. Spans question answering, knowledge-grounded dialogue, and summarization with hallucinated and non-hallucinated responses.

RAGTruth

~18,000 naturally generated RAG responses with human annotations at case and word level. Evaluates hallucinations specifically in retrieval-augmented settings.

FEVER

185,445 claims classified as Supported, Refuted, or NotEnoughInfo with evidence. Provides a clean fact-verification benchmark.

TruthfulQA

817 questions designed to test whether models reproduce common misconceptions. Serves as the out-of-domain test set.

Experimental Split

Split	Source	Purpose	Notes
Train / Validation	HaluEval and/or RAGTruth training portions	Model training and tuning	Primary supervised signal
Development	Held-out HaluEval / RAGTruth samples	Hyperparameter selection	In-distribution validation
Evaluation	RAGTruth test + FEVER + TruthfulQA	Final benchmark comparison	Cross-domain generalization
Manual Subset	Sampled end-to-end outputs	Correction quality + citation correctness	Human evaluation layer

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4) Methodology for Solutions

Pipeline Stages in Detail

1. Initial Generation

Instruction-tuned LLM generates answer without external verification (baseline)

2. Claim Extraction

Answer segmented into atomic or near-atomic factual claims

3. Hallucination Detection

Three complementary strategies: **self-consistency** (multiple responses, semantic/NLI agreement), **evidence-based verification** (retrieve passages, determine support/refutation), **LLM-based verification** (verifier model assesses claim against evidence)

4. Retrieval

Dense retriever (Sentence-BERT / contriever-style) retrieves top-k evidence passages from corpus

5. Correction

Unsupported/contradicted claims removed, rewritten, or replaced using evidence; Chain-of-Verification procedure tested

6. Final Answer

Corrected claims composed into final answer with citations attached to relevant claims

7. Evaluation

Compare systems on hallucination detection, factuality, end-to-end QA, citation quality, and efficiency

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5) Evaluation & Ablations

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Evaluation Metrics

- Hallucination Detection: Accuracy, Precision, Recall, F1, AUROC, AUPRC
- Factuality: Claim-level support rate, FACTSCORE-style atomic precision
- End-to-End QA: Exact match / F1, human and LLM-assisted evaluation
- Citation Quality: Correctness and relevance of attached evidence citations
- Efficiency: Latency, model calls, token usage, inference cost

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Key Ablations

A. LLM only

B. LLM + RAG

C. LLM + self-consistency

D. LLM + verification

E. LLM + RAG + verification

F. LLM + RAG + verification + correction

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